

Micro I, class 20

Along with some examples of strategic form games, I introduced three solution concepts for a strategic form game, $G = (N, \{S_i\}_{i \in N}, \{\succsim_i\}_{i \in N})$. The first, *dominant strategy equilibrium* is a strategy profile in which each player's strategy is best for that player *for any* strategy profile of the remaining players. I imagine that you got the idea that I found it a compelling solution concept, but often does not exist. I then suggested another solution concept also based on the idea of domination, but this time, players simply avoid *strictly dominated strategies*, rather than choose dominant ones. The concept is aptly named *iterative deletion of strictly dominated strategies*, because it starts from the original game, then deletes *all* of *each* player's strictly dominated strategies; in the game with the smaller strategy set, it then deletes *all* of *each* player's strictly dominated strategies in that game; proceeding iteratively, the solution concept is the limit of this decreasing (wrt $\supseteq$) sequence of sets. It always nonempty in finite games, but sometimes equals the entire set S of strategy profiles. In between these two notions is *Nash equilibrium*, a profile of strategies with the property that, *given* the strategy of the other players in the profile, each player i 's element of the profile is best for that player.

I then illustrated the notion of Nash equilibrium for (a special case of) the *Hotelling* game. Two ice cream vendors whose retail price is regulated must choose their locations on a beach of unit length, with consumers uniformly distributed along it. Consumers simply go to the nearest vendor. If the vendors locate at the same spot, they split the business equally. In this case, each vendor wants to choose a location to maximize its market share. Each player's strategy set is $[0, 1]$, and player i 's preferences over strategy profiles are represented by $(i, j = 1, 2, j \neq i)$

$$u_i(s_1, s_2) = \begin{cases} s_i + \frac{1}{2}(s_j - s_i) & \text{if } s_j > s_i \\ 1 - s_i + \frac{1}{2}(s_i - s_j) & \text{if } s_i > s_j \\ 1/2 & \text{if } s_1 = s_2. \end{cases}$$

It's obvious that the only candidate Nash equilibria involves $s_1 = s_2$ (since each vendor prefers to be closer to the other to increase market share). But for any $s_1 = s_2 \neq 1/2$, either vendor prefers to move a little bit into the longer end of the beach, to get more than half the market. The unique Nash equilibrium of this game is $(\frac{1}{2}, \frac{1}{2})$. The strategy $s_i = 0$ is strictly dominated (by $s_i = 1/2$) and $s_i = 1$ is strongly dominated (by $s_i = 1/2$).

Next time I will introduce the *Cournot* model of oligopoly (identical products version); and introduce the notion of a *best-reply correspondence*: $r_i(s_{-i}) = \{s'_i \in S_i \mid (s'_i, s_{-i}) \succsim (s_i, s_{-i}) \text{ for every } s_i \in S_i\}$. Defining $r(s) = \prod_i r_i(s_{-i})$, a profile $s^* \in S$ is a Nash equilibrium if and only if $s^* \in r(s^*)$. I will use this fact to present a theorem on existence of a Nash equilibrium.